

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

27-12-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

PART - I												
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10	
	A.	A	D	A	C	B,C	A,C	A,C	A,B,D	A,D	A,C,D	
	Q.	11	12									
	A.	A,B,C	A,B,D									
SECTION-II	Q.	1	2	3	4	5	6					
	A.	4.50	1.50	5.00	8.00	2.40	14.64					

PART-2 : CHEMISTRY

PART E-CHEMISTRY											
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	A	C	B,C,D	A,B,D	A,C	A,C,D	A,B,C,D	A,C
	Q.	11	12								
	A.	A,D	A,C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	1.50	7.00	25.00	0.10	8.00	9.00				

PART-3 : MATHEMATICS

PART C - MATHEMATICS												
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10	
	A.	C	B	C	A	A,B,D	B,C,D	B,C,D	A,B,C	B,C,D	A,B,C	
	Q.	11	12									
	A.	A,B,D	A,B,C									
SECTION-II	Q.	1	2	3	4	5	6					
	A.	4.00	25.00	4.00	7.33 or 7.34	6.00	10.50					

(1001CJA100120015)

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ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,B,C	A,B,C,D	A,B,C,D	A,C	A,B,C,D	A,B,C	A,B,C,D	A	D
	Q.	11	12								
	A.	C	A								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	33.67	12.00	57.14	1.52to 1.58	297.00	154to162				

PART-2 : CHEMISTRY

PART 2 - CHEMISTRY											
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	D	A,B,D	B,C	B,D	A,B,C,D	A,B	A,C,D	B	A
	Q.	11	12								
	A.	B	D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	9.00	8.00	6.00	1.21 to 1.22	4.00	4.00				

PART-3 : MATHEMATICS

PART C: MATHEMATICS											
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C	B,C	A,D	A,B,C	D	D	D	B	A	C
	Q.	11	12								
	A.	D	C								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	540.00	32.00	1998.00	4.33 or 4.34	1.20	9.00				



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PAPER-1

PART-1 : PHYSICS

SOLUTION

SECTION-I

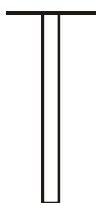
1. Ans. (A)

Sol. (I) Tension and stress are maximum at point of suspension

$$T = mg = \rho A \ell g$$

$$\frac{T}{A} = \rho \ell g = \sigma_B$$

$$\ell = \frac{\sigma_B}{\rho g}$$



2. Ans. (D)

Sol. $g_{\text{eff}} = g - \frac{g}{2} = \frac{g}{2}$

So $P_A + \rho(g_{\text{eff}})h = P_B$

$$P_B - P_A = \rho(g_{\text{eff}})h = \frac{\rho gh}{2}$$

3. Ans. (A)

Sol. For adiabatic expansion

$$T_i = 300 \text{ K}, V_i = V_2, T_f = ? V_f = 2\sqrt{2} V_2$$

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$$

$$300(V_2)^{2/3} = T_f(2\sqrt{2}V_2)^{2/3}$$

$$T_2 = \frac{300}{2} = 150 \text{ K}$$

4. Ans. (C)

Sol. $T \propto \frac{1}{V}$

$$T = k/V \text{ (k - constant)}$$

$$PV = nRT$$

$$PV = \frac{nRk}{V}$$

$$PV^2 = \text{constant}$$

$$C = \frac{R}{1-x} + \frac{fR}{2} = \frac{R}{1-2} + \frac{3R}{2} = -R + \frac{3R}{2}$$

$$C = \frac{R}{2}$$

5. Ans. (B,C)

Sol. Slope of curve $\propto \frac{1}{ms}$

$$\text{length of straight portion} \propto L$$

6. Ans. (A, C)

Sol. $\lambda_0 \rightarrow \frac{\lambda_0}{3}$

$$T' = 3T$$

$$P' = 81 P$$

7. Ans. (A,C)

Sol. $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$ & M is same

$$\therefore v_{\text{rms}} \text{ is same}$$

$$PV = nRT = \frac{N}{N_A} RT$$

$$\frac{P_1}{P_2} = \frac{N_1}{N_2} = \frac{1}{2}$$

8. Ans. (A,B,D)

Sol. For isochoric process

$$\Delta Q = nC_V \Delta T = n \left[\frac{3}{2} R \right] [\Delta T]$$

$$\& \Delta U = nC_V \Delta T = n \left[\frac{3}{2} R \right] [\Delta T]$$

$$W = 0$$

9. Ans. (A,D)

Sol. Line DB appear to pass through origin

$$P_0 = P_B - \text{Isobaric}$$

$$B \rightarrow C \text{ volume constant}$$

$$T \text{ decreasing } P\text{-decreasing}$$

$$P_C < P_B = P_D$$

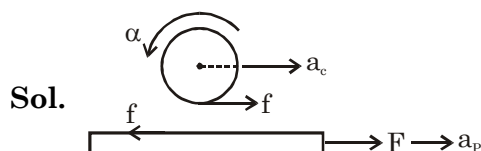
$$A \rightarrow B \text{ T-constant } V - \text{increasing}$$

$$P - \text{decreasing } P_B < P_A$$

10. Ans. (A,C,D)

Sol. As heat is transferred to block its temperature increases and temperature difference between both ends of rod decrease, Hence rate of flow of energy and rate of increase of temperature of block also decreases.

11. Ans. (A,B,C)



$$a_c + \alpha R = a_p$$

Friction force f is static

12. Ans. (A,B,D)

Sol. $\therefore V$ is increasing

So $W = +ve$

& A to B, P & V both are increasing

So T is also increasing

So $\Delta U \Rightarrow +ve$

So $\Delta Q = \Delta U + \omega$

$= +ve$

SECTION-II

1. Ans. 4.50

Sol. $W_{AB} = -2P_0 V_0$

$$W_{CD} = +12P_0 V_0$$

$$W_{EF} = -4P_0 V_0$$

$$W_{GH} = 3P_0 V_0$$

$$W_{BC} = W_{DE} = W_{RQ} = W_{HA} = 0$$

$$\text{Total work done} = 9P_0 V_0 = 2NP_0 V_0$$

$$N = 4.50$$

2. Ans. 1.50

Sol. $W_{\text{gas}} + W_{\text{spring}} + W_{\text{after}} = 0$

$$W_{\text{gas}} - \frac{1}{2} \times 200 \times 10^3 \times (0.2)^2 - 10^5 \times 0.05 \times$$

$$0.4 = 0$$

$$W_{\text{gas}} = 6000 \text{ J} = 6 \text{ kJ}$$

3. Ans. 5.00

Sol. $m_A = 2\text{kg}$ $m_B = 3\text{kg}$

$$\mathbf{v}_A = 4\hat{i} + 4.5\hat{j} - 2\hat{k} \quad \mathbf{v} = 0$$

$$2(4\hat{i} + 4.5\hat{j} - 2\hat{k}) + 3 \times 0 = 2(-2\hat{i} - 2\hat{k}) + 3(\bar{\mathbf{v}})$$

$$12\hat{i} + 9\hat{j} = 3\bar{\mathbf{v}}$$

$$\bar{\mathbf{v}} = 4\hat{i} + 3\hat{j} \quad |\mathbf{v}| = 5$$

4. Ans. 8.00

Sol. We have

$$V_{\text{flask}} = V_{\text{Hg}} + V_{\text{air}}$$

$$\text{As } (\Delta V)_{\text{air}} = 0$$

$$\therefore (\Delta V)_{\text{flask}} = (\Delta V)_{\text{Hg}}$$

$\therefore$ Cavity having same expansion co-efficient for material

$$\therefore V_f \gamma_f \Delta T = V_{\text{Hg}} \gamma_{\text{Hg}} \Delta T$$

$$\therefore \frac{V_f}{V_{\text{Hg}}} = \frac{\gamma_{\text{Hg}}}{\gamma_f} = \frac{1.8 \times 10^{-4}}{3 \times 9 \times 10^{-6}} = \frac{20}{3}$$

$$\therefore \text{Volume of flask} = 2000 \text{ cm}^3$$

5. Ans. 2.40

Sol. At n is total no. of molecule 20% are broken
Hence Now 0.8 N are diatomic and 0.4 N are monoatomic

Since volume is constant

$$\frac{P_1}{n_1 T_1} = \frac{P_2}{n_2 T_2}$$

$$\frac{P_1}{N \times 300} = \frac{P_2}{(1.2) \times 600}$$

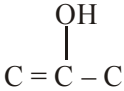
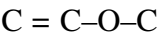
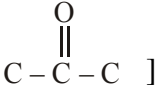
$$\frac{P_2}{P_1} = 1.2 \times 2 = 2.4$$

6. Ans. 14.64

Sol. Since fluid pressure is equal to blood pressure

$$\text{Hence } P_{\text{blood}} = P_0 + \rho gh$$

$$\text{Gauge pressure} = \rho gh = 1220 \times 10 \times 1.2 \\ = 14640 \text{ Pa} = 14.64 \text{ kPa}$$

PART-2 : CHEMISTRY	SOLUTION
SECTION-I 1. Ans. (D) 2. Ans. (B) 3. Ans. (A) 4. Ans. (C) Bulky group at equatorial position increases stability. 5. Ans. (B,C,D) 6. Ans. (A,B,D) 7. Ans. (A, C) 8. Ans. (A, C, D) $n(\text{FeSO}_4 \cdot 6\text{H}_2\text{O}) = \frac{2.6 \times 10^3}{260} = 10$ number of O-atom = $10 \times 10 \times N_A$ moles of H-atom = 10×12 molecules of H_2O = $10 \times 6 \times N_A$ moles of electron in SO_4^{2-} = $10 \times 1 \times 50$ 9. Ans. (A,B,C,D) 10. Ans. (A,C) 11. Ans. (A,D) 12. Ans. (A,C) SECTION-II 1. Ans. (1.50) 2. Ans. (7.00) 3. Ans. (25.00) 4. Ans. (0.10) Eq. of $\text{Na}_2\text{S}_2\text{O}_3 \Rightarrow$ Eq. of KBrO_3 $N \times 45 = \frac{125.25 \times 10^{-3}}{167} \times 6 \times 1000$ $N = 0.1$ $M = \frac{N}{n - \text{factor}} = 0.1M$	5. Ans.(8.00) 6. Ans.(9.00) Sol. $\text{C}-\text{C}-\text{CHO}$  $\text{HO}-\text{C}=\text{C}-\text{C}$  $\text{C}=\text{C}-\text{C}-\text{OH}$    $\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}$ 

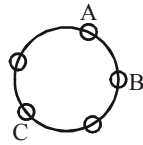
PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

First select two persons from A,B,C who wish to sit together in 3C_2 ways say A,B are together



then C must occupy the sit in one way & D,E can sit at remaining seats in 2 ways.

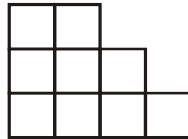
$$\Rightarrow \text{Total ways} = {}^3C_2 \times 1 \times 2 = 12 \text{ ways}$$

2. **Ans. (B)**

Case I :

Two rows contain 2 'X' each and one row has 1 X .

Possible ways =



$$4 \times {}^3C_2 + {}^4C_2 \times {}^3C_1 + {}^4C_2 \times {}^3C_2 \times 2 = 12 + 18 + 36 = 66$$

Case II :

One row has 3 'X' and two others have 1 X each

$$\text{ways} = {}^4C_3 \times {}^3C_1 \times {}^2C_1 + {}^4C_1 \times {}^2C_2 = 4 \times 3 \times 2 + 4 \times 2 = 24 + 8 = 32$$

Hence (B) is correct.

3. **Ans. (C)**

$$\frac{x^2}{4} + \frac{y^2}{9} = 1, P \equiv (2\cos\theta, 3\sin\theta).$$

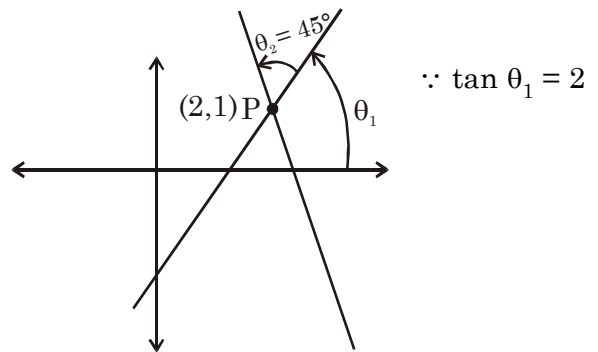
$$\therefore 4\cos^2\theta + 9\sin^2\theta = \frac{31}{4}.$$

$$\Rightarrow \cos^2\theta = \frac{1}{4} \Rightarrow \cos\theta = \pm \frac{1}{2}.$$

$$\therefore |12\cos\theta| = 6.$$

4. **Ans. (A)**

Sol.



let slope of new line is m

$$\therefore m = \tan(\theta_1 + \theta_2)$$

$$\Rightarrow m = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2}$$

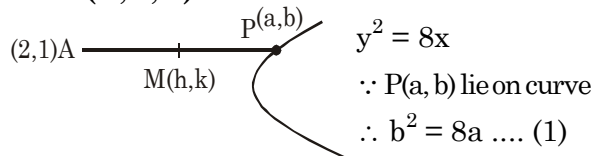
$$\Rightarrow m = \frac{2+1}{1-2 \cdot 1} = -3$$

$$\therefore \text{equation} \Rightarrow y - 1 = -3(x - 2)$$

$$\Rightarrow 3x + y = 7 \text{ Ans.}$$

5. **Ans. (A,B,D)**

Sol.



$$\therefore \frac{2+a}{2} = h \text{ \& \; } \frac{1+b}{2} = k$$

$$\Rightarrow a = 2h - 2; b = 2k - 1$$

put in (1)

$$\therefore (2k - 1)^2 = 8(2h - 2)$$

$$\Rightarrow (2y - 1)^2 = 16(x - 1) \text{ so } \lambda = 16$$

6. **Ans. (B,C,D)**

First select 6 places out of 11 for the boys in ${}^{11}C_6$ ways. Number of ways of arrangement at these places = 1 & at the remaining 5 places girls can stand in the way. Now there are 12 gaps & teachers can be arranged in ${}^{12}P_3$ ways

$$\therefore \text{total ways} = {}^{12}P_3 \times {}^{11}C_6 = 12 \times 11 \times 10 \times {}^{11}C_6 = 22 \times {}^{11}C_6 \times 60 \Rightarrow k = 60$$

7. Ans. (B,C,D)

Let the point be $(\alpha, \beta) \Rightarrow \beta = \alpha + 2$

Chord of contact of hyperbola is $T = 0$

$$\Rightarrow \frac{\alpha x}{6} - \frac{\beta y}{2} = 1 \Rightarrow \frac{\alpha x}{6} - \frac{(\alpha + 2)y}{2} = 1$$

$$\Rightarrow \alpha \left(\frac{x}{6} - \frac{y}{2} \right) - \left(\frac{2y}{2} + 1 \right) = 0$$

Since this passes through (λ, μ)

$$\therefore \frac{\lambda}{6} - \frac{\mu}{2} = 0 \text{ and } \frac{2\mu}{2} + 1 = 0 \Rightarrow \frac{\lambda}{\mu} = 3$$

8. Ans. (A,B,C)

Number of diagonals for n sides $= \frac{n(n-3)}{2}$

Number of diagonals for $(n+1)$ sides

$$= \frac{(n+1)(n-2)}{2}$$

$$\frac{(n+1)(n-2)}{2} - \frac{n(n-3)}{2} = 14$$

$$n = 10$$

(B) quantities are 1, 2, 3, 4

$$\frac{10!}{1! 2! 3! 4!}$$

$$(C) \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{(n-1)} - \frac{1}{n} \right) = 1 - \frac{1}{n}$$

9. Ans. (B,C,D)

Equation of hyperbola is $(x+2)(y-1) + k = 0$

$\therefore$ Hyperbola passes through $(1,0) \Rightarrow k = 3$

$\therefore$ Equation of hyperbola is $xy - x + 2y + 1 = 0$

10. Ans. (A,B,C)

(A) Tangent of slope m is $y = mx + \frac{3}{m}$

For $m = 3$, tangent is $y = 3x + 1$

(B) Normal of slope m is $y = mx - 6m - 3m^3$

For $m = 2$, normal is $y = 2x - 36$

(C) Directrix is $x = -3$

Any point on it is $(-3, k)$

Its chord of contact is $ky = 6(x-3)$

$$\text{or } 6x - ky = 18 \text{ or } 2x - \frac{k}{3}y = 6$$

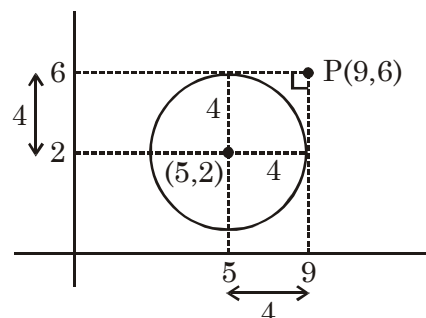
(D) Line $x - 2y + 12 = 0$ is a tangent to

$$y^2 = 12x$$

at $(12, 12)$.

11. Ans. (A,B,D)

Sol.



12. Ans. (A,B,C)

$$\frac{(x-1)^2}{25} - \frac{(y-1)^2}{16} = 1$$

$$\therefore \text{eccentricity} = \sqrt{1 + \frac{16}{25}} = \frac{\sqrt{41}}{5}$$

$$\text{directrix, } x = \pm \frac{a}{e} \Rightarrow x = \pm \frac{25}{\sqrt{41}} + 1$$

Foci $x = \pm ae, y = 0$

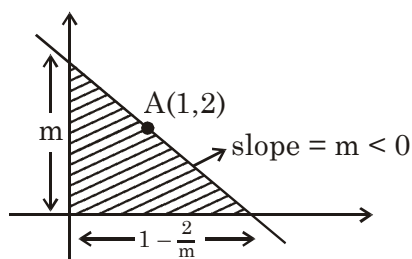
$$\Rightarrow x = \pm \sqrt{41} + 1, y = 1$$

$$\text{Latus rectum} = \frac{2b^2}{a} = \frac{32}{5}$$

SECTION-II

1. Ans. 4.00

Sol. $\therefore (2k+1)x + (3k-2)y + (3-8k) = 0$
 $\Rightarrow (x-2y+3) + k(2x+3y-8) = 0$
 $\Rightarrow L_1 + \lambda L_2 = 0$



$\therefore \left. \begin{array}{l} L_1 : x-2y+3=0 \\ \& L_2 : 2x+3y-8=0 \end{array} \right\} \text{fixed point : (1, 2)}$

equation of line : $y-2 = m(x-1)$

$\therefore$ X-intercept $= 1 - \frac{2}{m}$ & Y-intercept $= 2 - m$

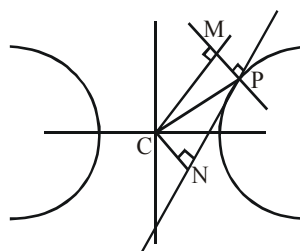
Area $= \frac{1}{2}(2-m)\left(1 - \frac{2}{m}\right)$; $m < 0$

Area $= \frac{1}{2}\left[4 + (-m) + \left(\frac{-4}{m}\right)\right]$

$\left\{ \begin{array}{l} \because (-m) \text{ is +ve} \\ \text{Now, A.M.} \geq \text{G.M.} \\ \Rightarrow \frac{(-m) + \left(\frac{-4}{m}\right)}{2} \geq \sqrt{(-m) \cdot \left(\frac{-4}{m}\right)} \\ \Rightarrow (-m) + \left(\frac{-4}{m}\right) \geq 4 \end{array} \right\}$

$\therefore (\text{Area})_{\text{least}} = \frac{1}{2}[4 + 4] = 4 \text{ Ans.}$

2. Ans. 25.00



Equation of tangent at P is

$3\sec\theta \cdot x - 4\tan\theta \cdot y - 12 = 0$

$\therefore CN = \frac{12}{\sqrt{9\sec^2\theta + 16\tan^2\theta}}$

Equation of normal at P is

$4 \times \cos\theta + 3y \cot\theta = 25$

$\therefore CM = \frac{25}{\sqrt{16\cos^2\theta + 9\cot^2\theta}}$

$\therefore A = \frac{1}{2} \cdot CM \times CN$

$= \frac{1}{2} \cdot \frac{25 \cdot 12}{\sqrt{144 + 81\operatorname{cosec}^2\theta + 256\sin^2\theta + 144}}$

but $\frac{81\operatorname{cosec}^2\theta + 256\sin^2\theta}{2} \geq \sqrt{81\operatorname{cosec}^2\theta \cdot 256\sin^2\theta}$

$= 9 \cdot 16 = 144$

$\therefore$ Max. value of $81 \operatorname{cosec}^2\theta + 256 \sin^2\theta = 2 \cdot 144$

$\therefore A_{\text{max.}} = \frac{1}{2} \cdot \frac{25 \cdot 12}{\sqrt{4 \cdot 144}} = \frac{1}{2} \cdot \frac{25 \cdot 12}{2 \cdot 12} = \frac{25}{4}$

$\therefore 4A = 25$

3. Ans. 4.00

$\frac{2}{t_1} \times \frac{2}{t_2} = -1$

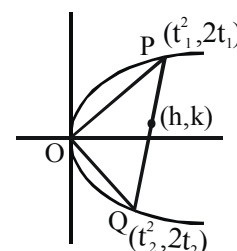
$\Rightarrow t_1 t_2 = -4$

$h = \frac{t_1^2 + t_2^2}{2}$ & $k = t_1 + t_2$

$\Rightarrow 2h = (t_1 + t_2)^2 - 2t_1 t_2$

$\Rightarrow 2h = k^2 + 8 \Rightarrow y^2 = 2(x - 4)$

$\therefore \ell = 4$



4. Ans. 7.33 or 7.34

Sol. $(2,3)A$ $B(1,-2)$

$4x - y = 10$

$P(a, 4a - 10)$

$\therefore |PA - PB| < AB \quad \{\because \text{In } \triangle ABP\}$

$\Rightarrow |PA - PB|_{\max} = AB \quad \{\text{when } A, B, P \text{ are collinear}\}$

$(2,3)A$ $B(1,-2)$

$4x - y = 10$

$P(a, 4a - 10)$

$\therefore m_{AB} = m_{AP} \quad \therefore P(-3, -22)$

$\Rightarrow \frac{3+2}{2-1} = \frac{13-4a}{2-a} \Rightarrow a = -3 \quad \Rightarrow p = -3 \text{ \& } q = -22$

so $\left| \frac{q}{p} \right| = \left| \frac{-22}{-3} \right| = 7.33 \text{ or } 7.34$

5. Ans. 6.00

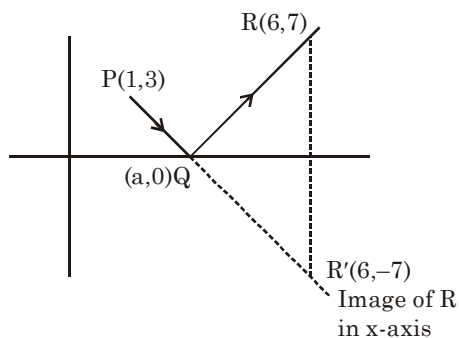
$$4(x^2 + 2x + 1) + 9(y^2 - 4y + 4) = 36$$

$$\frac{(x+1)^2}{9} + \frac{(y-2)^2}{4} = 1$$

Ellipse with major axis as the maximum chord, having the length is 6.

6. Ans. 10.50

Sol.



$\therefore P, Q, R'$ are collinear

$$\therefore m_{PQ} = m_{PR'}$$

$$\Rightarrow \frac{3}{1-a} = \frac{10}{-5}$$

$$\Rightarrow a = \frac{5}{2}$$

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (B,C,D)

$$\text{Sol. } [E] = [k\sqrt{a}] = \left[\sqrt{\frac{P^4}{2m^2}} \right]$$

k is dimensionless so

$$[E] = [\sqrt{a}] \quad (\text{B is correct})$$

and

$$[\sqrt{a}] = \left[\sqrt{\frac{P^4}{2m^2}} \right]$$

$$\text{so } [a] = \left[\frac{P^4}{2m^2} \right]$$

$$[P^4] = [2m^2a]$$

$$[P] = [(m^2a)^{1/4}] \quad (\text{C is correct})$$

and

$$[\sqrt{a}] = [E]$$

$$[a] = [E^2] \Rightarrow \left[\frac{E^2}{a} \right] = [1] \text{ dimensionless}$$

2. Ans. (A,B,C)

Sol. $\vec{A} + \vec{D} = \vec{R}$ (1) (triangle law of add.)

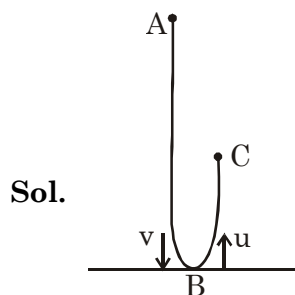
$$\vec{B} + \vec{C} = \vec{R} \text{(2) triangle law of add.)}$$

from (1) & (2)

$$\vec{A} + \vec{D} = \vec{B} + \vec{C} \Rightarrow \vec{A} - \vec{B} = \vec{C} - \vec{D} \quad (\text{B correct})$$

$$\text{from (1) } \vec{A} = \vec{R} - \vec{D}$$

3. Ans. (A, B, C, D)



A to B

$$s = ut + \frac{1}{2}at^2$$

$$v = u + at$$

$$20 = 0 + \frac{1}{2} \times 10t^2$$

$$v = 0 + 10 \times 2$$

$$t = 2 \text{ sec}$$

$$v = 20 \text{ m/s}$$

B to C

$$v^2 = u^2 + 2as$$

$$0 = u^2 - 2 \times 10 \times 5$$

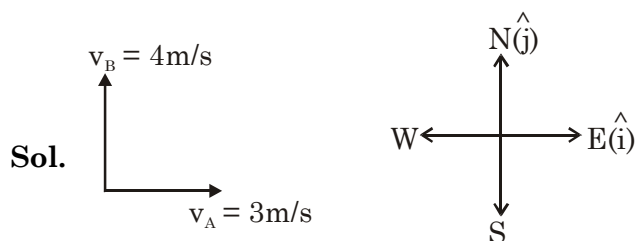
$$u = 10 \text{ m/s}$$

$$v = u + at$$

$$0 = 10 - 10t$$

$$t = 1 \text{ sec}$$

4. Ans. (A, B, C, D)



$$\vec{v}_A = 3\hat{i}; \vec{v}_B = 4\hat{j}$$

$$\vec{v}_{A/B} = 3\hat{i} - 4\hat{j} \Rightarrow |\vec{v}_{AB}| = \sqrt{3^2 + 4^2} = 5 \text{ m/s}$$

in 3 sec.

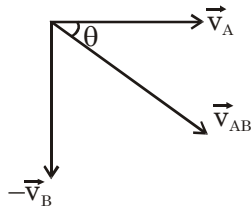
$$\vec{S}_A = (3\hat{i}) \times 3 = 9\hat{i} \text{ m}$$

$$\vec{S}_B = (4\hat{j}) \times 3 = 12\hat{j} \text{ m}$$

$$|\vec{S}_{AB}| = |9\hat{i} - 12\hat{j}| = 15m$$

$$\tan\theta = \frac{4}{3}$$

$$\theta = 53^\circ$$

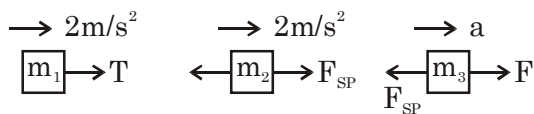


5. **Ans. (A, C)**

Sol. Value of maximum friction force on 10 kg block is $= f_{\max} = \mu (10g)$
so when $mg > 10\mu g$
system moves otherwise it does not move.

6. **Ans. (A,B,C,D)**

Sol. m_1 & m_2 are connected with strings so will have same acceleration.



$$T = m_1(2) = 2N$$

$$F_{SP} - T = m_2(2) \Rightarrow F_{SP} = 4 + 2 = 6N$$

$$F - F_{SP} = m_3 a \Rightarrow a = \frac{18-6}{3} = 4 \text{ m/s}^2$$

7. **Ans. (A,B,C)**

Sol. $Y = \frac{\frac{F}{A}}{\frac{\Delta L}{L}} \Rightarrow \Delta L = \frac{FL}{AY}$

Elastic potential energy

$$= \frac{1}{2} \times (\text{stress})^2 \times \frac{\text{volume}}{Y}$$

$$= \frac{1}{2} \left(\frac{F}{A} \right)^2 \times \frac{AL}{Y} = \frac{F^2 L}{2AY}$$

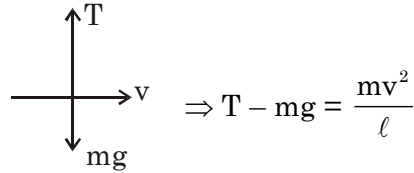
8. **Ans. (A, B, C, D)**

Sol. If $0 \leq v < \sqrt{2g\ell}$ remain between B & C

$v = \sqrt{2g\ell}$ just reaches to point C

$\sqrt{2g\ell} \leq v < \sqrt{5g\ell}$ string slacks between C & D

$v \geq \sqrt{5g\ell}$ it completes vertical circle at lowest point



$$T = mg + \frac{m}{\ell}(2g\ell) = 3mg$$

9. **Ans. (A)**

Sol. (i) $\Delta U = \frac{f}{2} nR\Delta T = \frac{f}{2} (P_2 V_2 - P_1 V_1)$

$$\Delta U = \frac{3}{2} (2P_0(2V_0) - (2P_0 V_0)) = 3P_0 V_0 \text{ (positive)}$$

(ii) D $\rightarrow$ A

$$P_B V_B = P_C V_C$$

$$2P_0 \times 2V_0 = P_C \times 3V_0$$

$$P_C = \frac{4}{3} P_0 = P_B$$

(iii) $\Delta T = 0, \Delta U = 0$

(iv) C $\rightarrow$ D (isobaric)

$$\Delta Q = nC_p \Delta T$$

$$\Delta Q = \frac{5}{2} \left(\frac{4}{3} P_0 V_0 - 4P_0 V_0 \right) = -\frac{20}{3} P_0 V_0$$

10. **Ans. (D)**

Sol. $v_{\text{rms}} = \sqrt{\frac{3RT}{M_w}} = \sqrt{\frac{3PV}{nM_w}}$

$n = \text{constant}$

$M_w = \text{constant}$

So ratio of v_{rms} at any state is ratio of

$\sqrt{PV}$ at that state

$P = \text{pressure}$

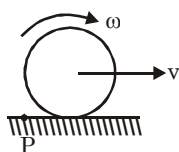
$v = \text{volume}$

$$\text{efficiency} = \frac{\text{Work total}}{\Delta Q \text{ exclusive}}$$

11. Ans. (C)

12. Ans. (A)

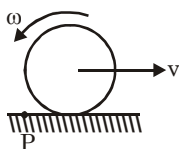
Sol. Case-I



about point P angular momentum will be conserved as not torque on the disc is zero.

$$L_P = mvR + \frac{mR^2}{2}\omega \text{ (clock wise)}$$

Case-II



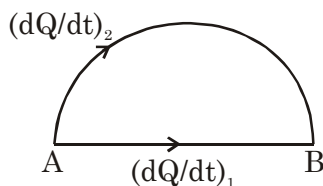
$$L_P = mvR - \frac{mR^2}{2}\omega \text{ (clockwise)}$$

If angular momentum comes out clockwise then it moves in the same direction of initial velocity if it is zero then finally it stops and if it is anticlockwise then it moves in left direction.

SECTION-II

1. Ans. 33.67

Sol.



$$\left(\frac{dQ}{dt}\right)_1 = \frac{kA\Delta T}{2R} ; \left(\frac{dQ}{dt}\right)_2 = \frac{kA\Delta T}{\pi R}$$

$$\frac{dQ}{dt} = \left(\frac{dQ}{dt}\right)_1 + \left(\frac{dQ}{dt}\right)_2 = \frac{kA\Delta T}{R} \left(\frac{1}{2} + \frac{1}{\pi}\right)$$

$$\left(\frac{dQ}{dt}\right) = \frac{18000 \times 2 \times 10^{-4} \times (100 - 20)}{\frac{14}{2}}$$

$$\left(0.5 + \frac{1}{3.14}\right) = 33.67 \text{ J/sec}$$

2. Ans. 12.00

$$\text{Sol. } \frac{\Delta v}{v} = \gamma \Delta T$$

$$\beta = \frac{\Delta P}{\frac{\Delta v}{v}} \Rightarrow \Delta P = \beta(\gamma \Delta T) = 10^4 \times 3 \times 2 \times 10^{-5} \times$$

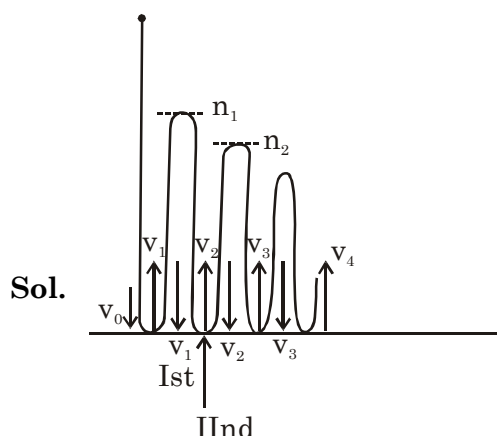
$$20 = 12 \text{ Pa}$$

3. Ans. 57.14

$$\text{Sol. Rate of heat given} = \left(\frac{dm}{dt}\right) S \Delta T$$

$$= 4 \times 1 \times 60 = 240 \text{ cal/s} = 57.14 \text{ J/s}$$

4. Ans. 1.52 to 1.58



Sol.

$$v_1 = ev_0 \text{ (velocity after 1st collision)}$$

$$v_2 = e^2v_0 \text{ (velocity after 2nd collision)}$$

$$v_3 = e^3v_0 \text{ (velocity after 3rd collision)}$$

$$v_4 = e^4v_0 \text{ (velocity after 4th collision)}$$

$$v_0 = \sqrt{2gH}$$

$$v_1 = \sqrt{2gh_1} \text{ (} h_1 \rightarrow \text{maximum height after 1st collision on)}$$

$$\sqrt{2gh_1} = e\sqrt{2gH}$$

$$h_1 = e^2H$$

$$100 = 125 e^2 \Rightarrow e = \sqrt{\frac{100}{125}} = \frac{10}{5\sqrt{5}}$$

$$e = \frac{2}{\sqrt{5}}$$

so $v_2 = \sqrt{2gh_2}$ and

$$e^2 v_0 = \sqrt{2gh_2} \quad v_4 = \sqrt{2gh_4}$$

$$\sqrt{2gh_2} = e^2 \sqrt{2gH} \quad e^4 v_0 = \sqrt{2gh_4}$$

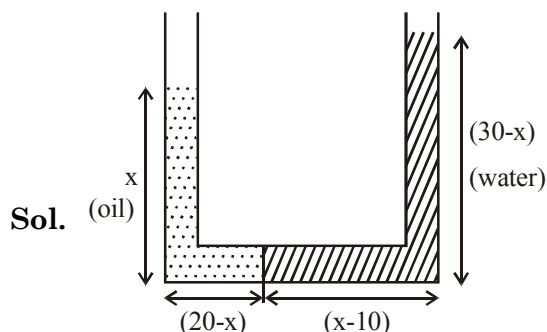
$$h_2 = e^4 H \dots\dots(1) \quad h_4 = e^8 H \dots\dots(2)$$

$$\frac{h_2}{h_4} = \frac{1}{e^4} = \left(\frac{\sqrt{5}}{2}\right)^4 = \frac{25}{16} = 1.562$$

5. Ans. 297.00

Sol. $w_2 - w_1 = mg\Delta h$
 $= (99 \times 10 \times 1) \times 10^{-3} \times 10 \times 0.3$
 $= 9.9 \times 0.3 = 2.97 \text{ J}$

6. Ans. 1.54 to 1.62



Sol.

$$\begin{aligned} \therefore \rho_{\text{oil}} gx + \rho_{\text{oil}} (20 - x)a + \rho_{\text{water}} (x - 10)a &= \rho_{\text{water}} g (30 - x) \\ 800 \times 10x + 800(20 - x) \times 5 + 1000(x - 10)(5) &= 1000 \times 10(30 - x) \\ 8x + 80 - 4x + 5x - 50 &= 300 - 10x \\ 19x = 270 \Rightarrow x &= \frac{270}{19} = 14.21 \text{ cm} \\ \therefore \Delta H = (30 - x) - x &= (30 - 2x) = 1.58 \end{aligned}$$

PART-2 : CHEMISTRY

SOLUTION

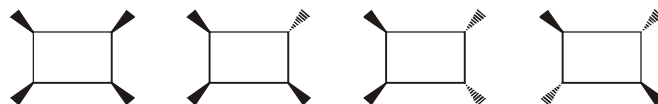
SECTION-I

1. Ans. (A,B,C)
2. Ans. (D)
3. Ans. (A,B,D)
4. Ans. (B,C)
5. Ans. (B,D)
6. Ans. (A,B,C,D)
7. Ans. (A,B)
8. Ans. (A,C,D)
9. Ans. (B)
10. Ans. (A)
11. Ans. (B)
12. Ans. (D)

SECTION-II

1. Ans. (9.00)
2. Ans. (8.00)
3. Ans. (6.00)
4. Ans. (1.217) Range (1.21 to 1.22)
5. Ans. (4.00)
6. Ans. (4.00)

Sol. 4 Geometrical isomers



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (B,C)**

Sol. We have $\alpha + \beta = -a$ (1)

$$\alpha\beta = a + 1 \quad \text{....(2)}$$

$\therefore$ On adding (1) and (2), we get

$$\alpha + \beta + \alpha\beta = 2$$

$$(\alpha + 1)(\beta + 1) = 2$$

since a and b are integers

$$\Rightarrow \text{either } \alpha + 1 = 1 \text{ and } \beta + 1 = 2$$

$$\text{or } \alpha + 1 = -1 \text{ and } \beta + 1 = -2$$

hence $\alpha = 0$ and $\beta = 1$ or $\alpha = -2$ and $\beta = -3$

$$\Rightarrow a = -1 \text{ or } 5$$

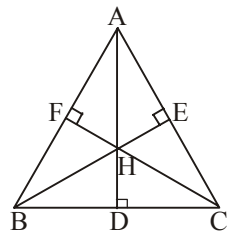
2. **Ans. (B,C)**

Sol. We have $\frac{HB \times HC \times BC}{\text{Area}(\Delta HBC)}$

$$= \frac{HB \times HC \times BC}{\frac{1}{2}(HB)(HC) \sin A} = \frac{2a}{\sin A} = \frac{(2a)(2R)}{a}$$

$$= 4R = \frac{abc}{\Delta} \quad \text{....(1)}$$

$$s = \frac{a+b+c}{2} = 7$$



$$\text{Now } \Delta = \sqrt{7 \times 4 \times 2 \times 1} = \sqrt{56}$$

From equation (1), we get

$$4R = \frac{3 \times 5 \times 6}{\sqrt{56}} = \frac{45}{\sqrt{14}} = \frac{m}{\sqrt{n}}$$

$$m = 45 \text{ and } n = 14$$

Hence $a + b = 59$ **Ans.**

3. **Ans. (A,D)**

Sol. The condition of concurrency is

$$\begin{vmatrix} a & 4 & a^2 \\ 1 & -2 & -6 \\ 3 & 1 & -4 \end{vmatrix} = 0$$

$\therefore$ On expanding along R_1 , we get

$$a^2 + 2a - 8 = 0$$

$$\Rightarrow (a + 4)(a - 2) = 0$$

Hence $a = -4, 2$

4. **Ans. (A,B,C)**

Sol. $y^2 = 4(x - 1)x^2 + y^2 = 1$

$$\text{tangent } y = m(x - 1) + \frac{1}{m}$$

for circle = $\perp^{\text{ar}}$ from $(0, 0) = 1$

$$\left| \frac{-m + \frac{1}{m}}{\sqrt{m^2 + 1}} \right| = 1$$

$$\frac{(1 - m^2)^2}{m^2} = (m^2 + 1)$$

$$1 + m^4 - 2m^2 = m^4 + m^2$$

$$m \rightarrow \infty \quad 3m^2 = 1$$

$$x = 1 \quad m = \pm 1/\sqrt{3}$$

$$m = \frac{1}{\sqrt{3}}$$

$$m = -\frac{1}{\sqrt{3}}$$

$$y = \frac{1}{\sqrt{3}}(x - 1) + \sqrt{3}$$

$$\sqrt{3}y + x + 2 = 0$$

5. **Ans. (D)**

Sol. We know $a + b = 18$ and $ac = 72$.

For the first equation, the only possible (a, b) are $(8, 10)$ and $(10, 8)$ because the number are distinct.

For the second equation, the only possible (a, c) are $(8, 9)$ and $(9, 8)$ because the value of a is the same in both, we get $a = 8$, $b = 10$ and $c = 9$.

The harmonic mean of b and

$$c = \frac{2}{\frac{1}{10} + \frac{1}{9}} = \frac{180}{19} \text{ **Ans.**}$$

6. Ans. (D)

Sol. $y = \begin{vmatrix} x & 1 & -3 \\ 2 & 2 & 1 \\ \frac{1}{x-1} & 0 & \frac{1}{2} \end{vmatrix}$

$$= x(1-0) - \left(1 - \frac{1}{x-1}\right) - \frac{3}{2} \left(0 - \frac{2}{x-1}\right)$$

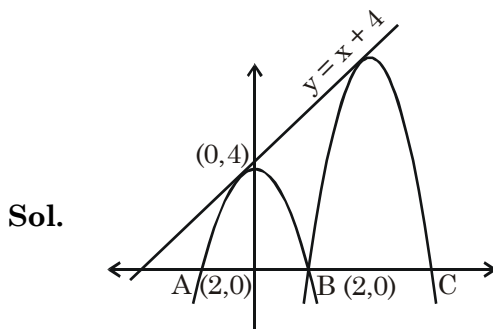
$$= x - 1 + \frac{1}{x-1} + \frac{3}{x-1} = (x-1) + \frac{4}{(x-1)}$$

By applying A.M.-G.M. inequality, we have

$$y = (x-1) + \frac{4}{x-1} \geq 2\sqrt{(x-1)\frac{4}{x-1}} = 4$$

(As $x > 1$) Ans.

7. Ans. (D)



$$(x - \alpha)^2 = -(y - (\alpha + 4))$$

The curve passes through (2, 0)

$$(2 - \alpha)^2 = -(0 - (\alpha + 4))$$

$$\alpha^2 - 5\alpha = 0 \Rightarrow \alpha = 0 \text{ or } \alpha = 5$$

$$(x - 5)^2 = -(y - 9) \text{ put } y = 0 \Rightarrow x = 2, 8$$

8. Ans. (B)

Sol. $C_{______} = 4! = 24$

$$D_{______} = 4! = 24$$

$$M_{______} = 4! = 24$$

$$\textcircled{S} C_{______} = 3! = 6$$

$$\textcircled{S} D_{______} = 3! = 6$$

$$\textcircled{S} \textcircled{M} \textcircled{C} D W = 1$$

$$\textcircled{S} \textcircled{M} \textcircled{C} W D = 1$$

Solution 9 and 10

9. Ans. (A)

10. Ans. (C)

Sol. (I)

$$x = \frac{2009 \times 2008 \times 2007 + 1}{2008 \times 2007 + 2007} = 2008 + \frac{1}{2009 \times 2007}$$

$$[x] - 2008 = 0 \quad (I) \rightarrow P$$

$$(II) P(\sqrt{6} \cos \theta, \sqrt{2} \sin \theta)$$

$$6 \cos^2 \theta + 2 \sin^2 \theta = 2^2$$

$$4 \cos^2 \theta = 2$$

$$\cos^2 \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$[\theta] = 0 \quad 2 \quad 3 \quad 5$$

$$(II) \rightarrow P, R, S, U$$

$$(III) \frac{1}{e^2} + \frac{1}{(e')^2} = 1 \Rightarrow e' = \frac{5}{4}$$

$$4e' = 5 \quad (III) \rightarrow (U)$$

$$(IV) \text{ Locus of } P \text{ is directrix } x + 1 = 0$$

$$\text{hence distance} = 3 \quad (IV) \rightarrow (S)$$

Solution 11 and 12

11. Ans. (D)

12. Ans. (C)

Sol. (I) We have $\cos \theta = \frac{a}{b+c}$ (Given)

$$\text{Now } \sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{a^2}{(b+c)^2}}$$

$$= \sqrt{\frac{(b+c)^2 - a^2}{(b+c)^2}}$$

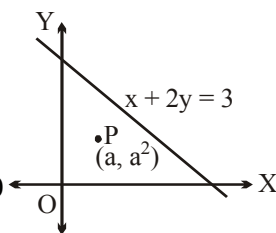
$$= \sqrt{\frac{(b+c+a)(b+c-a)}{(b+c)^2}} = \sqrt{\frac{2S(2S-2a)bc}{bc(b+c)^2}}$$

$$\sin \theta = \frac{2\sqrt{bc}}{b+c} \cos \frac{A}{2} \Rightarrow k = 2$$

(II) For point (a, a^2) to lie inside the triangle must satisfy

$$a > 0 \quad \dots (1)$$

$$a^2 > 0 \quad \dots (2)$$



$$\text{and } a + 2a^2 - 3 < 0$$

$$\Rightarrow (2a + 3)(a - 1) < 0$$

$$\Rightarrow (a - 1) < 0 \quad (\because 2a + 3 > 0)$$

$$\Rightarrow a < 1 \quad \therefore a \in (0, 1)$$

$\therefore$ No integral value of a exist

(III) $3^{x-1} + 3^{x-2} + 3^{x-3} + \dots = 2(5^2 + 5 + 1 + \dots)$

$$\frac{3^{x-1}}{1 - \frac{1}{3}} = 2 \left[\frac{25}{1 - \frac{1}{5}} \right] \Rightarrow \frac{3^x}{2} = \frac{2[5 \times 25]}{4}$$

$$3^x = 5^3$$

$$a = 3 \log_x 5$$

So $N = 5$ **Ans.**

Hence three lines are possible.

(IV) Equation of line passing through $(3, 6)$ of slope m is

$$(y - 6) = m(x - 3) \dots (1)$$

$$\text{Put } y = 0 \text{ in equation (1), we get } \left(3 - \frac{6}{m}, 0 \right)$$

And put $x = 0$ in equation (1), we get $(0, 6 - 3m)$

$$\text{Now } \frac{1}{2} \left| 3 - \frac{6}{m} \right| |6 - 3m| = 36$$

$$\Rightarrow (m - 2)^2 = 8 |m|$$

$$\therefore \text{ Either } m^2 - 4m + 4 = 8m \text{ or } m^2 - 4m + 4 = -8m$$

$\therefore$ Three distinct values of m are possible.

Hence three lines are possible.

SECTION-II

1. **Ans. 540.00**

Sol. Let $\alpha_1 = A$, $\beta_1 = AR$, $\alpha_2 = AR^2$, $\beta_2 = AR^3$

we have $\alpha_1 + \beta_1 = 6$

$$\Rightarrow A(1 + R) = 6 \quad \dots (1)$$

$$\alpha_1 \beta_1 = p$$

$$\Rightarrow A^2 R = p \quad \dots (2)$$

$$\text{Also } \alpha_2 + \beta_2 = 54$$

$$\Rightarrow AR^2(1 + R) = 54 \quad \dots (3)$$

$$\alpha_2 \beta_2 = q$$

$$A^2 R^5 = q \quad \dots (4)$$

Now, on dividing equation (3) by equation (1), we get

$$\frac{AR^2(1+R)}{A(1+R)} = \frac{54}{6} = 9$$

$$\Rightarrow R^2 = 9$$

$\therefore R = 3$ (As it is an increasing G.P.)

$\therefore$ On putting $R = 3$ in equation (1)

.....

$$A = \frac{6}{4} = \frac{3}{2}$$

$$\therefore p = A^2 R = \frac{9}{4} \times 3 = \frac{27}{4}$$

$$\text{and } q = A^2 R^5 = \frac{9}{4} \times 243 = \frac{2187}{4}$$

$$\text{Hence } q - p = \frac{2187 - 27}{4} = \frac{2160}{4} = 540$$

2. **Ans. 32.00**

Sol. Radius of given circle

$$= \sqrt{4 + 2 - c} = \sqrt{6 - c} = a \text{ (say)}$$

Now radius of circle $S_1 = \frac{a}{\sqrt{2}}$, radius of

circle $S_2 = \frac{a}{2}$ and so on.

$$\text{Now } a + \frac{a}{\sqrt{2}} + \frac{a}{2} + \dots \infty = 2 \text{ (Given)}$$

$$\Rightarrow a \left(\frac{1}{1 - \frac{1}{\sqrt{2}}} \right) = 2 \Rightarrow \frac{a\sqrt{2}}{\sqrt{2}-1} = 2$$

$$\Rightarrow a = 2 - \sqrt{2} = \sqrt{6-c}$$

$$\Rightarrow 4 + 2 - 4\sqrt{2} = 6 - c.$$

$$\text{Hence } c = 4\sqrt{2} \text{ Ans.}$$

3. **Ans. 1998.00**

Sol. Total 3 digit numbers = ${}^5C_3 \times 3! = 60$

Sum of each column = $12[1 + 2 + 3 + 4 + 5]$

$$= 180$$

19	18	
×	×	×
×	×	×
×	×	×
199	18	0

$$\left. \vphantom{\begin{array}{c} 19 \\ \times \\ \times \\ \times \\ 199 \end{array}} \right\} 60$$

Hence sum = $1998 \times 10 \Rightarrow N = 1998$ **Ans.**

4. **Ans. 4.33 or 4.34**

Sol. $b^4 = 3$;

$$\text{hence } \log_{b^2} 27 = \frac{3a}{2}$$

$$\Rightarrow 3 \log_{b^2} 3 = \frac{3a}{2}$$

$$\Rightarrow \log_{\sqrt{3}} 3 = \frac{a}{2} \Rightarrow a = 4$$

$$\therefore a^2 - b^4 = 16 - 3 = 13 \text{ Ans.}$$

5. **Ans. 1.20**

$$(a-1)^2 + 16 = (a-9)^2$$

$$a^2 - 2a + 1 + 16 = a^2 + 81 - 18a$$

$$16a = 64 \Rightarrow a = 4$$

$$(b-4)^2 + 1 = 81 + b^2$$

$$\Rightarrow b^2 - 8b + 17 = 81 + b^2$$

$$\Rightarrow 8b = -64$$

$$\Rightarrow b = -8$$

$$\therefore a - b = 12 \text{ ans.}$$

6. **Ans. 9.00**

Sol. ${}^6C_4 - {}^4C_2 = 9$